H

RAFFLES INSTITUTION

2014 Year 6 Preliminary Examination
Higher 2

CANDIDATE NAME								
CIVICS GROUP	1	4	S	0	3		INDEX NUMBER	

BIOLOGY

Paper 2 Core Paper

9648/02
15th SEPTEMBER 2014
2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **ONE** question.

At the end of the examination, **hand in your essay SEPARATELY.**
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/14
3	/11
4	/11
5	/ 7
6	/12
7	/13
Section B	
8 or 9	/20
Total	/100

This document consists of **19** printed pages.

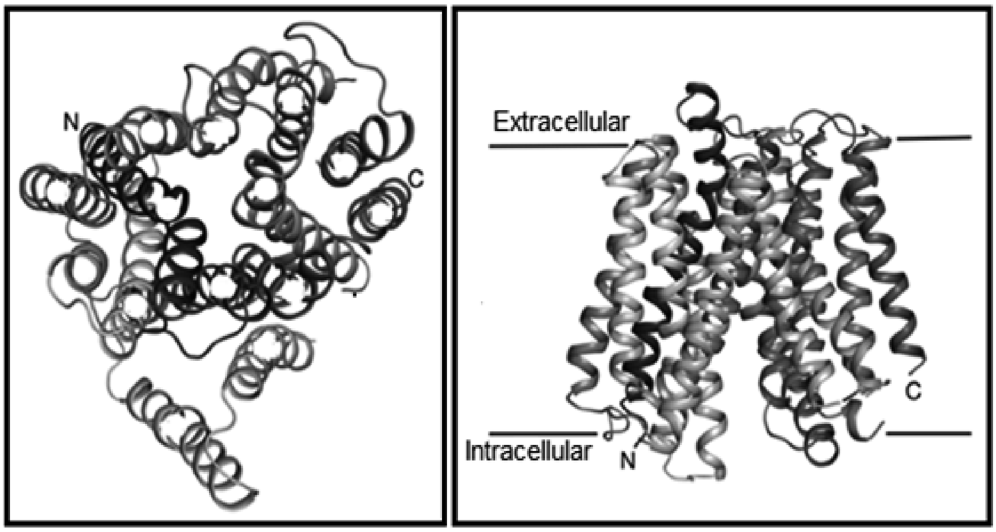
For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Differences in ion concentrations are maintained across cell membranes. These differences are significant in nervous transmission.

Fig. 1.1(a) shows a top-view diagram of the potassium ion leak channel protein and Fig. 1.1(b) shows its relative position on the cell surface membrane.



(a) Topview (b) Sideview

Fig. 1.1

(a) With reference to Fig. 1.1 and your knowledge on membrane protein structure, describe one key structural feature of this transport protein that allows it to perform its function.

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(b) The resting membrane potential of neurons is maintained at -70mV.
Describe how the resting membrane potential is maintained across cell membranes.

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For Examiner's Use

(c) Serotonin is a neurotransmitter that improves the mood of a person by stimulating the pleasure centre of the brain. Monoamine oxidase is an enzyme that breaks down serotonin. Moclobemide is an inhibitor of monoamine oxidase and was developed as an antidepressant. Fig. 1.2 below shows the effect of moclobemide on the rate of the enzyme-catalyzed reactions.

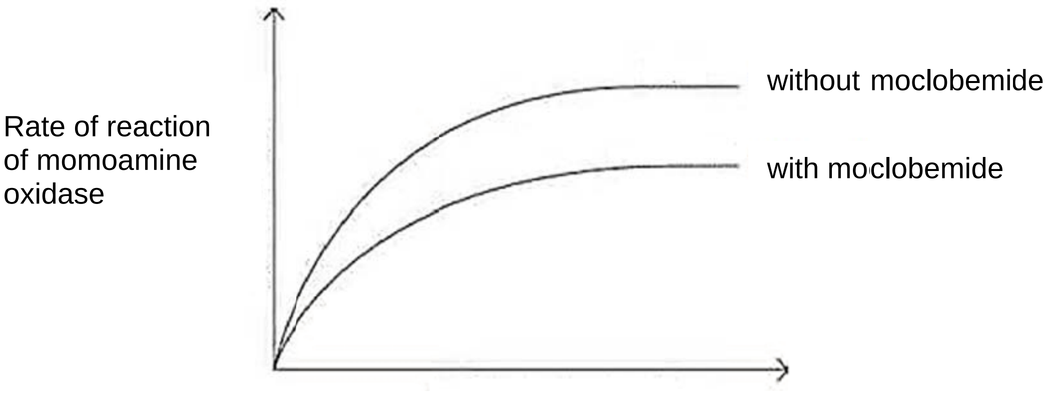

Fig. 1.2

(i) Explain the effect of moclobemide on the rate of reaction of monoamine oxidase.

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(ii) Moclobemide acts as an antidepressant. Explain its effect at the synapse.

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[Total : 12]

For Examiner's Use

2 Telomeres are found at the ends of linear chromosomes.

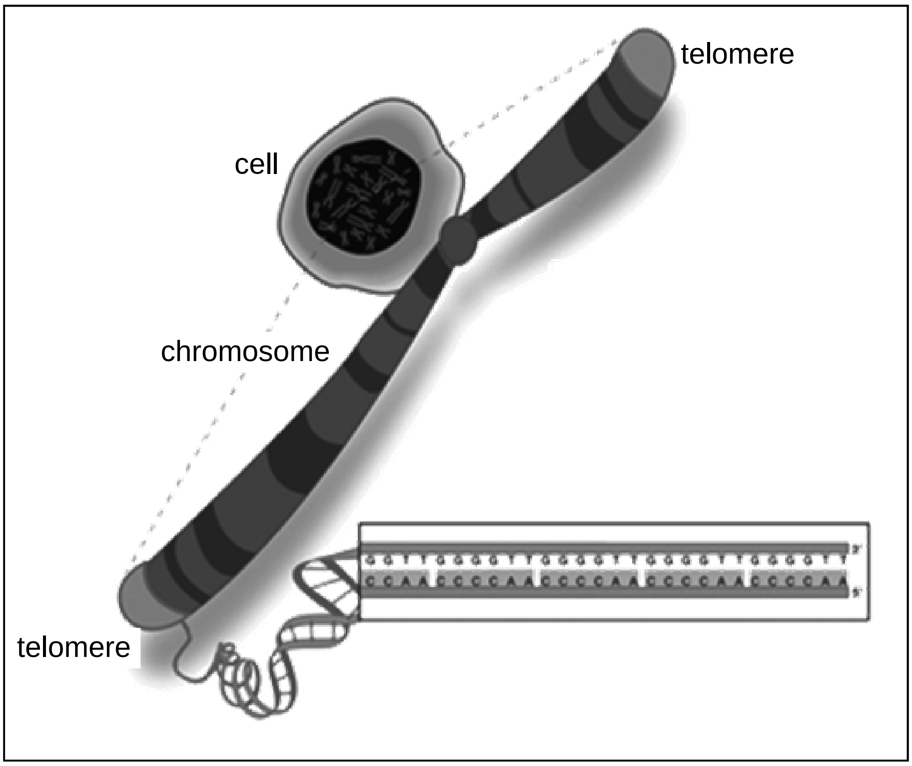

Fig. 2.1

(a) State one structural feature of telomeres and relate this structure to its function in the cell.

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……. [2]

Activation of telomerase and subsequent telomere maintenance are often observed in cancer cells.

Fig. 2.2 shows the regulation of the expression of the human telomerase reverse transcriptase (*hTERT*) gene that encodes the catalytic subunit of telomerase.

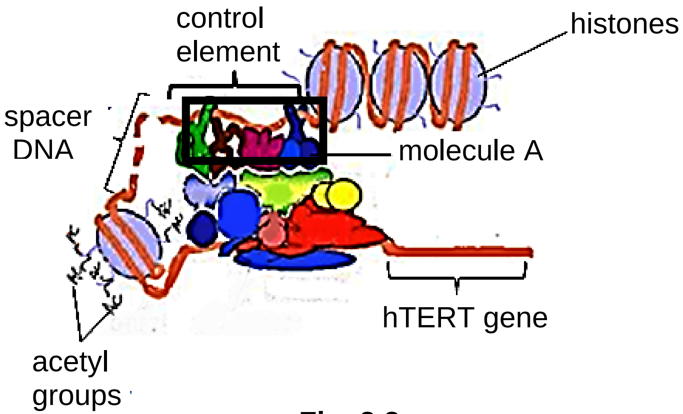

Fig. 2.2

(b) With reference to Fig 2.2,

(i) describe how molecule A is involved in the regulation of hTERT gene expression.

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.... [2]

(ii) describe one other way that upregulation of hTERT gene expression is achieved.

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.... [2]

(c) Describe the function of hTERT protein in the cell.

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.... [3]

For Examiner's Use

(d) Telomerase RNA is encoded by a separate gene (hTR). Describe the role of telomerase RNA in the cell.

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(e) Fig. 2.3 shows the drug Imetelstat which is used to treat cancer cells with high telomerase activity. Imetelstat is an oligonucleotide with a covalently-bound hydrocarbon tail.

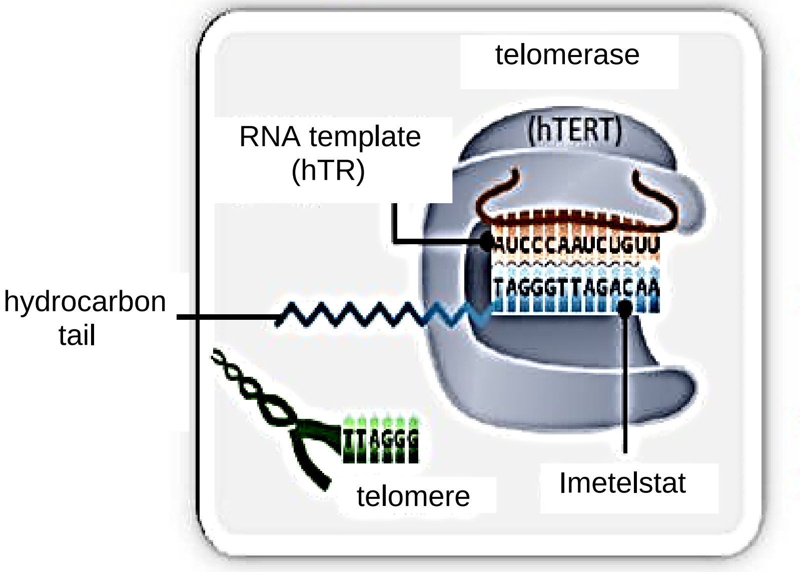

Fig. 2.3

(i) With reference to Fig. 2.3, describe the mode of action of Imetelstat on telomerase.

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(ii) Suggest a reason for the presence of the covalently-bound hydrocarbon tail on Imetelstat.

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[Total : 14]

3 A student identified cell A as a prokaryotic cell as it does not contain a membrane-bound nucleus and cell B as a eukaryotic cell as it contains a membrane-bound nucleus.

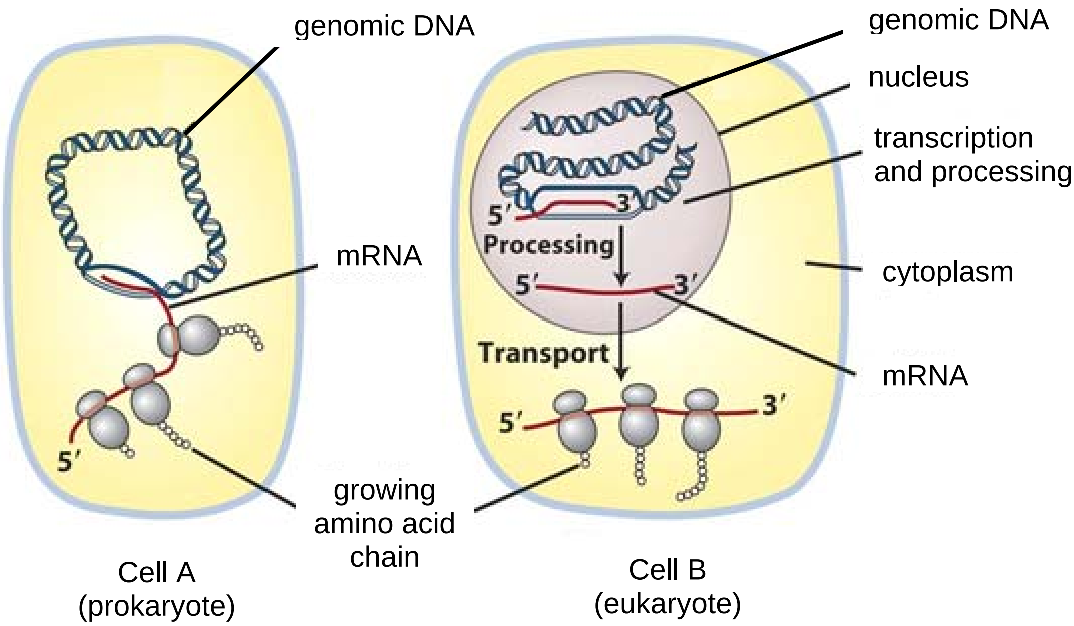

Fig. 3.1

(a) With reference to the processes occurring within the cells shown in Fig. 3.1, give two other reasons why cell A is likely to be a prokaryotic cell.

Reason 1: ………………………………………………………………………………………………

……[1]

Reason 2: ………………………………………………………………………………………………

……[1]

(b) Tetracycline is known to prevent bacterial growth by interfering with protein synthesis.

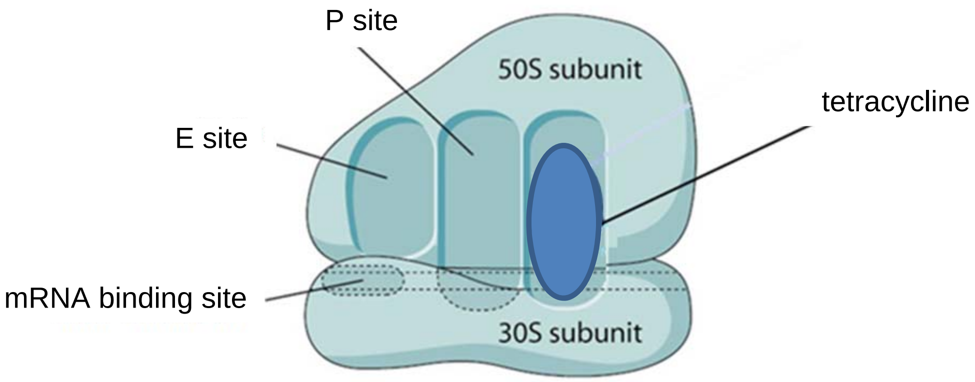

Fig. 3.2

(i) With reference to Fig. 3.2, explain how tetracycline prevents protein synthesis.

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.... [3]

(ii) Suggest why tetracycline has no effect on human cells.

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.... [2]

(iii) Furvina® is an antibiotic that inhibits initiation of translation. Indicate with an "X" on Fig. 3.2, the region where Furvina® binds. [1]

For Examiner's Use

(c) *Vibrio cholera* is a bacterium that produces a cholera toxin that causes severe diarrhoea. The gene for the toxin is unique to *Vibrio cholera* and is not found in human cells.

In a study to investigate potential drugs for treating cholera, antisense DNA oligonucleotides that are complementary to the cholera mRNA were used to inhibit the expression of the toxin. This mechanism is summarized in Fig.3.3.

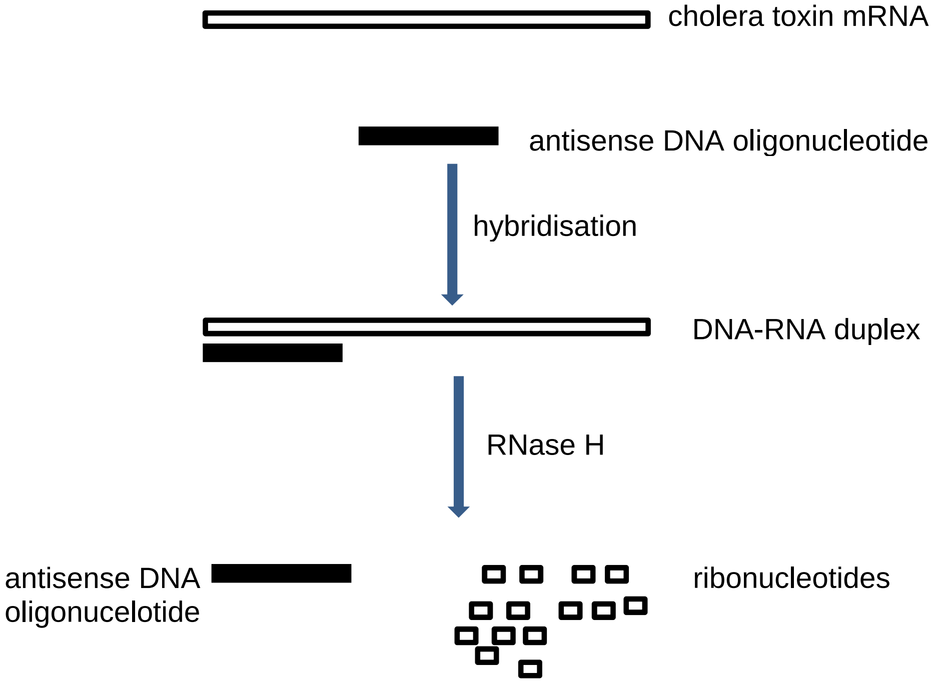

Fig. 3.3

(i) In Fig. 3.3, RNase H destroys the mRNA strand, and both the RNase H enzyme and the antisense DNA oligonucleotide remains intact at the end of the process.

With reference to Fig 3.3, describe how protein synthesis is prevented by the antisense DNA strand.

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Fig. 3.4 shows a segment of an antisense DNA oligonucleotide that contains modified nucleotides.

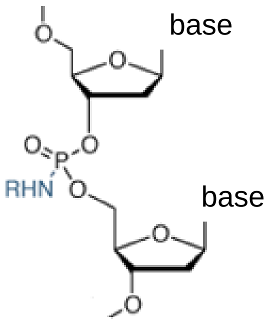

Fig. 3.4

(ii) With reference to Fig. 3.4, suggest why this anti-sense DNA strand does not get degraded by DNases.

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[Total : 11]

4 Fig. 4.1 shows a propagator which scientists use to investigate the effect of light intensity on the rate of photosynthesis in lettuce.

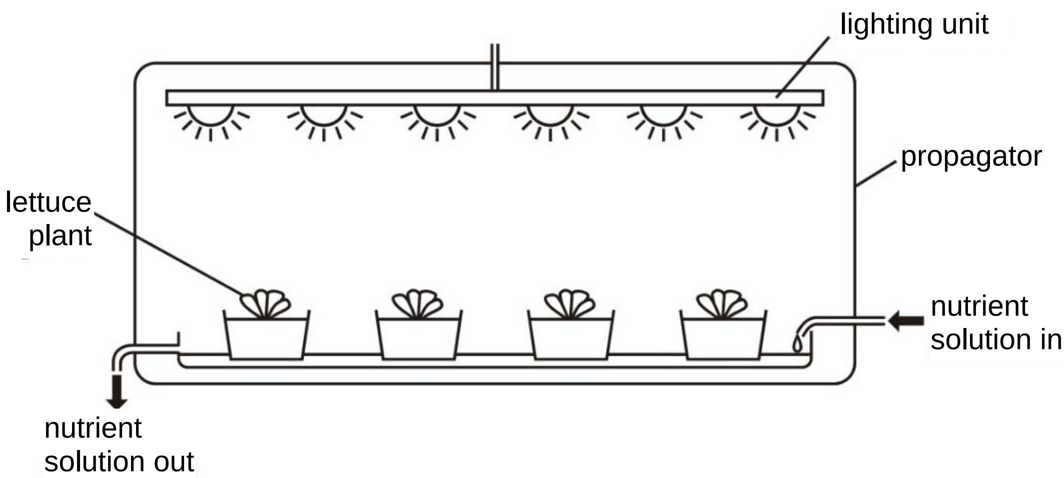

Fig. 4.1

(a) Explain why the temperature has to be kept constant throughout the above experiment.

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For Examiner's Use

(b) A similar experiment was conducted to investigate how various factors affect the rate of photosynthesis in lettuce. Fig. 4.2 below shows the results of the experiments conducted.

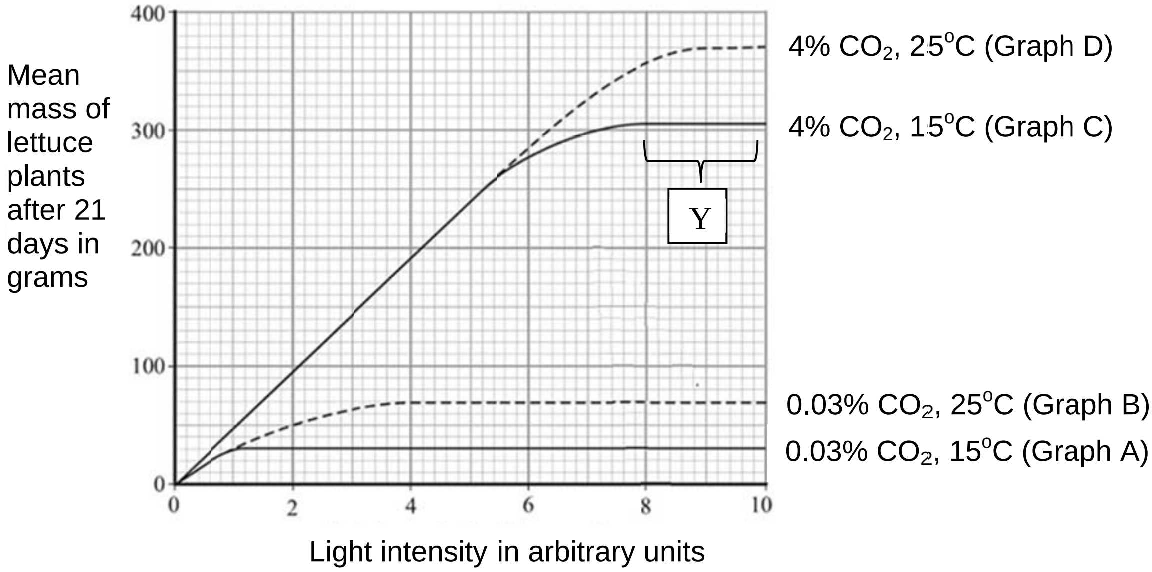

Fig. 4.2

(i) With reference to Fig. 4.2, state the best conditions for the growth of lettuce.

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(ii) Explain the region marked Y.

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(iii) Describe and explain the effect of increasing carbon dioxide concentration on the mean mass of lettuce after 21 days at 25ºC.

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……… [4]

(iv) The average carbon dioxide content of the atmosphere is 0.035%. Using this fact, and the information given in Fig. 4.2, what conclusion can be made about how carbon dioxide affects photosynthesis in the natural environment?

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……… [2]

[Total : 11]

5 The rabies virus is classified in the same group as the influenza virus. Fig. 5.1 shows a simplified rabies virus life cycle in an infected cell.

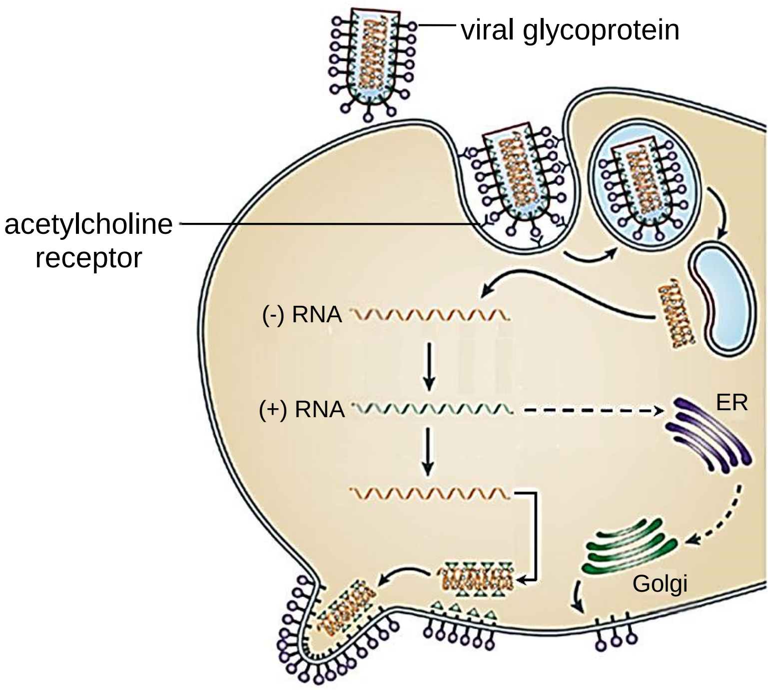

Fig. 5.1

(a) With reference to Fig. 5.1, identify the type of host cell that the rabies virus infects.

……… [1]

For Examiner's Use

(b) **(i)** Using the information given in Fig.5.1, explain the role of a named viral enzyme in the rabies reproductive cycle.

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(ii) Using Fig. 5.1, distinguish between the reproductive cycles of lambda phage and rabies virus. [3]

stage	**lambda phage**	**rabies virus**
penetration		
integration		
release		

[Total : 7]

6 Two pure bred varieties of a rice plant, Y and Z have been developed by the Raffles Rice Research Institute (R^3I). Variety Y plants with heights ranging from 21-25 cm, have round leaves and are glyphosate resistant. Variety Z plants with heights ranging from 56-60cm, have wrinkled leaves and are glyphosate sensitive.

When variety Z was pollinated by variety Y, all its progeny are glyphosate resistant and round leaves. The height distribution of the F_1 generation are shown in the Table 6.1 below:

height/cm	**number of F_1 plants**
26-30	1
31-35	10
36-40	54
41-45	51
46-50	8
51-55	3

Table 6.1

Table 6.2 shows the phenotypes of the F_2 progeny obtained when the F_1 plants were self-pollinated.

height/cm	**number of F_2 progeny**			
	round leaves & glyphosate resistant	**wrinkled leaves & glyphosate resistant**	**round leaf & glyphosate sensitive**	**wrinkled leaf & glyphosate sensitive**
21-25	15	1	0	5
26-30	88	2	2	28
31-35	114	4	3	35
36-40	152	5	5	48
41-45	152	5	6	47
45-50	116	4	4	36
51-55	86	2	3	27
56-60	13	0	0	4
Sub-total	**736**	**23**	**23**	**230**

Table 6.2

(a) State whether the alleles that code for round leaves and glyphosate resistance are dominant or recessive. Explain your answer.

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(b) With reference to the data in Table 6.2, state the number of genes and alleles that control the response of the rice plants to glyphosate. Explain your answer.

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(c) In cats, the gene controlling coat colour is sex-linked. The coat of male cats are either black or ginger while those of females are black, ginger or tortoiseshell (patchwork pattern of black, ginger and cream).

In a litter of kittens found in the basement of Raffles Rice Institute, half the number of females were tortoiseshell and the other half were black; half the number of males were black and the other half were ginger. Using a genetic diagram, show how mating between two parents of different genotype and phenotype, results in the phenotypic ratio as observed in this litter of kittens. [4]

(d) It is often difficult to determine if coat colour is autosomal or sex-linked. Describe a way to determine if coat colour is autosomal or sex-linked.

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[Total : 12]

7 Evolutionary biologists apply the fact that molecular evolution occurs in a clock-like fashion to construct phylogenetic trees.

A scientist studied the number of nucleotide substitutions in the DNA coding for the alpha globin chain of haemoglobin, in a group of related birds, A, B, C and D.

The nucleotide sequences of the alpha globin chain of birds were then compared. Fig. 7.1 shows the results of the comparison between birds A and B, C and D, A and C, and, B and D.

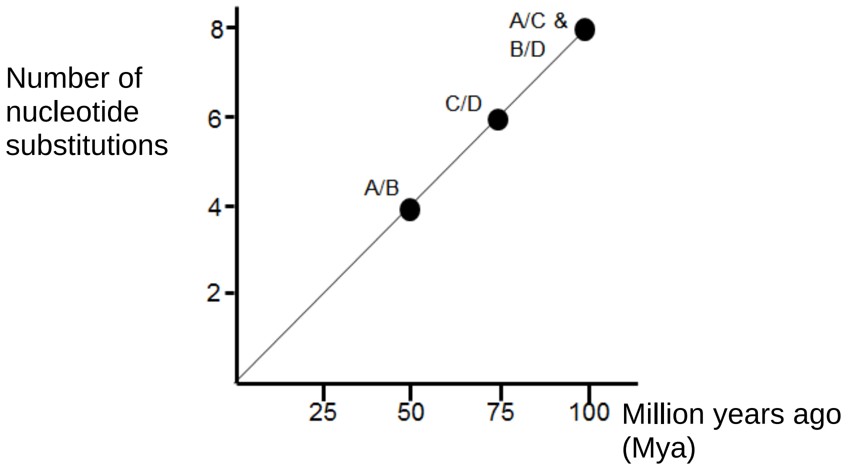
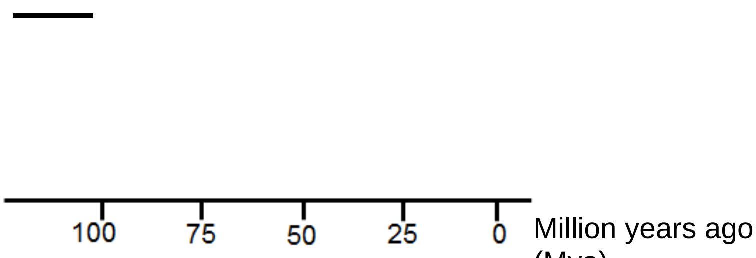


Fig. 7.1

(a) **(i)** Based on the information given in Fig. 7.1, draw a phylogram representing the 4 species of birds in the space below. [2]



(ii) Explain why the scientist chose to study nucleotide substitutions instead of other types of mutations when constructing the molecular clock?

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(iii) Using the information in Fig.7.1, calculate the rate of mutation of the alpha globin chain.

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In the Hawaiian archipelago, there exists a group of endemic birds called the Hawaiian honeycreepers. At least 56 species of Hawaiian honeycreepers are known to have existed, although all but 18 of them are now extinct (latest extinction as recently as 2004). Fig. 7.2 shows the phylogenetic tree of the Hawaiian honeycreeper species. The 4 grey vertical bars represent the time of the formation of the 4 Hawaiian islands.

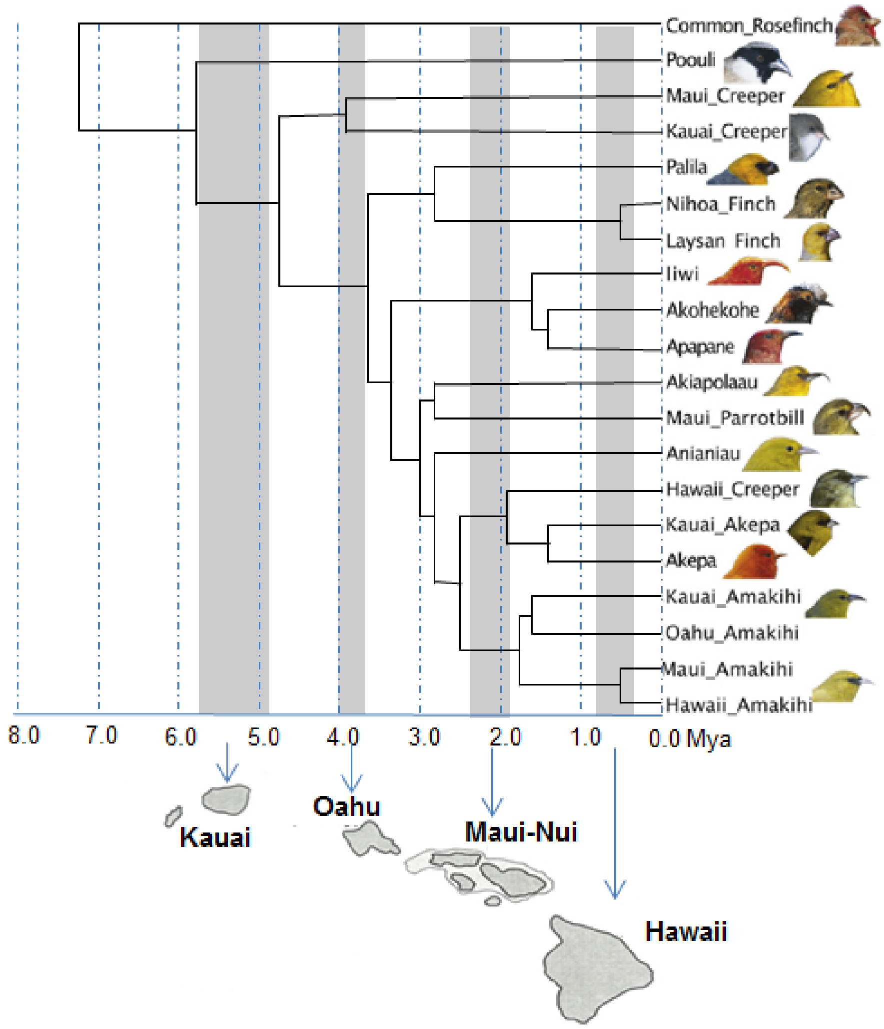

Fig. 7.2

For Examiner's Use

(b) Explain how the common ancestor of Hawaiian honeycreepers evolved into so many different species in just under 6 million years?

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(c) With reference to Fig.7.2, name the island whose emergence corresponds to the greatest number of speciation events?

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(d) The Maui Creeper, as its name implies, is endemic to Maui-Nui, yet it appeared sometime during the formation of Oahu. Explain its current distribution.

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…………………………………………………………………………………………………….. [2]

[Total: 13]

Section B
Answer EITHER 8 OR 9.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Describe the role of meiosis in natural selection. [8]

(b) Compare oxidative phosphorylation with photophosphorylation. [6]

(c) Describe how cellular responses during cell signaling can be terminated. [6]

[Total: 20]

9 **(a)** Discuss the significance of NAD and NADP. [8]

(b) Compare the differences between the structures of cellulose and insulin. [4]

(c) Outline the role of proteins in the movement of molecules in and out of a cell. [8]

[Total: 20]

End of Paper